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Tool Used: Elastic Security (Kibana), Security Onion

1. Objective

The objective of this practical is to perform advanced log analysis by applying log correlation, anomaly detection, and log enrichment techniques using Elastic Security. This helps in identifying suspicious activities and improving threat detection capabilities.

2. Tools Required

- Elastic Security (Kibana)
- Security Onion (optional)
- Google Sheets (for documentation)
- Boss of the SOC Dataset (sample logs)

3. Log Correlation

3.1 Aim

To correlate failed login attempts (Event ID 4625) with suspicious outbound network activity.

3.2 Procedure

1. Open Kibana dashboard (http://localhost:5601)
2. Navigate to Discover section
3. Upload sample logs (JSON/LOG format)
4. Apply filter:
   event.code: 4625
5. Identify repeated failed login attempts
6. Search for outbound traffic using:
   destination.ip: *
7. Compare timestamps and source IPs
8. Identify patterns indicating suspicious behavior

3.3 Observation Table

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request
2025-08-18 12:02	4624	192.168.1.100	8.8.8.8	Possible compromise

3.4 Result

The correlation of failed login attempts followed by outbound traffic indicates a possible brute-force attack leading to unauthorized access and data exfiltration.

4. Anomaly Detection

4.1 Aim

To detect unusual high-volume data transfers that may indicate data exfiltration.

4.2 Procedure

1. Go to Security → Rules → Create Rule
2. Select "Custom Query Rule"
3. Enter query:
   network.bytes_out > 1000000
4. Set time window to 1 minute
5. Select appropriate index (logs)
6. Save and enable rule
7. Simulate or upload data with high traffic

4.3 Observation

- Alerts triggered when outgoing data exceeds threshold
- Identified abnormal spikes in network usage

4.4 Result

The rule successfully detected high-volume data transfers, helping identify potential data exfiltration attempts.

5. Log Enrichment (GeoIP)

5.1 Aim

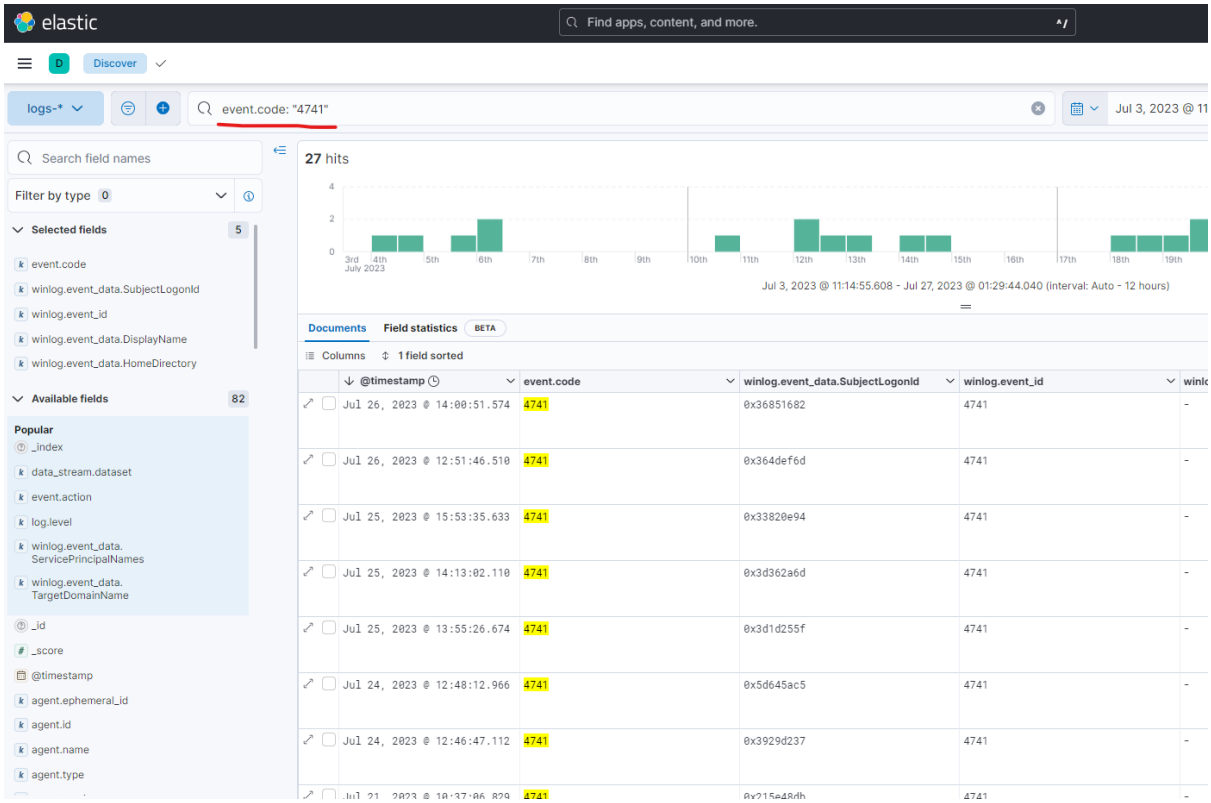
To enrich log data with geographical information for better analysis.

5.2 Procedure

1. Navigate to Ingest Pipelines in Kibana
2. Create or edit pipeline
3. Add GeoIP processor
4. Select field: source.ip
5. Apply pipeline to logs
6. View enriched logs in Discover or Maps

5.3 Observation

- Logs now include country, city, and location details
- Easier identification of suspicious foreign IPs



Elastic

Search Elastic

Discover

New Save Open Share Inspect

Search KQL Last 15 minutes Show dates Refresh

+ Add filter

kublr_demo*

Search field names

Filter by type 0

Selected fields

_source

Available fields

_id
_index
_score
_type
@timestamp
@version
annotations.authentication.k8s.io/legacy-token
annotations.authorization.k8s.io/decision
annotations.authorization.k8s.io/reason
annotations.k8s.io/deprecated
annotations.k8s.io/removed-release
apiVersion
auditID

107,044 hits

Apr 6, 2022 @ 18:06:49.790 - Apr 6, 2022 @ 18:21:49.790 Auto

Count

@timestamp per 30 seconds

Time	_source
Apr 6, 2022 @ 18:21:46.363	apiVersion: audit.k8s.io/v1 headers.request_path: / headers.http_user_agent: Go-http-client/1.1 headers.content_type: application/json headers.http_accept: - headers.content_length: 64255 headers.http_version: HTTP/1.1 headers.accept_encoding: gzip headers.http_host: kublr-logging-logstash.kublr:8080 headers.request_method: POST level: Request objectRef.resourceVersion: 232019576 objectRef.uid: 3e08b913-28c6-4c68-942d-7019282b8572 objectRef.apiGroup: coordination.k8s.io objectRef.apiVersion: v1 objectRef.namespace: azure-arc-k8s-ext
Apr 6, 2022 @ 18:21:46.363	apiVersion: audit.k8s.io/v1 headers.request_path: / headers.http_user_agent: Go-http-client/1.1 headers.content_type: application/json headers.http_accept: - headers.content_length: 64255 headers.http_version: HTTP/1.1 headers.accept_encoding: gzip headers.http_host: kublr-logging-logstash.kublr:8080 headers.request_method: POST level: Request objectRef.resourceVersion: 232019542 objectRef.uid: 30a89543-d076-4b42-a579-0c9e91454be6 objectRef.apiGroup: keda.sh objectRef.apiVersion: v1alpha1 objectRef.namespace: azure-arc-k8s-ext
Apr 6, 2022 @ 18:21:46.363	apiVersion: audit.k8s.io/v1 headers.request_path: / headers.http_user_agent: Go-http-client/1.1 headers.content_type: application/json headers.http_accept: - headers.content_length: 64255

Recent cases Created by anyone

Suspicious emails reported
Several employees have reported receiving emails from unknown addresses.
2 days ago 1

test
test
2 days ago 0

View all cases

Recent timelines

You haven't favorited any timelines yet. Get out there and start threat hunting!

View all timelines

Security news

Empowering security teams with prebuilt protections
2023-05-11
Want to know more about the prebuilt detection rules and security content and why it matters to you? Read on!

Elastic Security Labs unpacks ICEDID
2023-05-05
A comprehensive tutorial with Elastic Security Labs open source tools.

Elastic Security Labs discovers the LOBSHOT malware
2023-04-25
An analysis of LOBSHOT, an hVNC malware family spreading through Google Ads.

Elastic Security Labs outlines an

Alert trend

Stack by kibana.alert.rule.name View alerts

Showing: 1,544 alerts

Malware Prevention Alert
sample

Events

Stack by event.dataset View events

Showing: 12,491,108 events

Other
elastic_agent.filebeat
elastic_agent.metricbeat
dns
socket

Host events

View hosts

Showing: 4,028,963 events

Auditbeat	8,762
Endpoint Security	3,954,881
Filebeat	55,937
Winlogbeat	9,383

Network events

View network

Showing: 5,888,408 events

Auditbeat	363,400
Filebeat	0
Packetbeat	5,525,008

Discover

New Open Share Alerts Inspect Save

Kibana Sample Data Logs

Filter your data using KQL syntax

Last 7 days

Search field names 0

Available fields 28

@timestamp
agent
bytes
bytes_counter
bytes_gauge
clientip
event.dataset
extension
geo.coordinates
geo.dest
geo.src
geo.srcdest
host
hour_of_day
index
ip
machine.os
machine.ram
memory
message
phpmemory
referer

Auto interval No breakdown

Jul 29, 2024 @ 00:00:00.000 - Aug 5, 2024 @ 13:49:42.975 (interval: Auto - 3 hours)

Documents (1,731) Patterns (4) Field statistics

message

Count	Log rate	Examples	Actions
606		162.127.63.105 - - [2018-07-23T05:23:29.090Z] "GET /beats/filebeat/filebeat-6.3.2-linux-x86.tar.gz HTTP/1.1" 200 9805 "-" "Mozilla/5.0 (X11; Linux x86_64 162.127.63.105 - - [2018-07-23T05:23:29.090Z] "GET /beats/filebeat/filebeat-6.3.2-linux-x86.tar.gz HTTP/1.1" 200 9805 "-" "Mozilla/5.0 (X11; Linux x86_64 162.127.63.105 - - [2018-07-23T05:23:29.090Z] "GET /beats/filebeat/filebeat-6.3.2-linux-x86.tar.gz HTTP/1.1" 200 9805 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:6.0a1) Gecko/20110421 Firefox/6.0a1"	
592		27.239.147.47 - - [2018-07-23T00:00:00.000Z] "GET /apm-server/apm-server-6.3.2-windows-x86.zip HTTP/1.1" 200 9424 "-" "Mozilla/5.0 (X11; Linux i686) AppleWebKit/534.24 (KHTML, like Gecko) Chrome/11.0.696.50 Safari/534.24"	
490		217.31.123.207 - - [2018-07-23T02:48:18.823Z] "GET /beats HTTP/1.1" 200 7624 "-" "Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1; .NET CLR 1.1.4322)"	
43		184.0.152.233 - - [2018-07-23T07:21:43.501Z] "GET /beats HTTP/1.1" 503 0 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:6.0a1) Gecko/20110421 Firefox/6.0a1"	

Rules

Create new rule

Rule preview

1 Define rule

Rule type

Custom query — Use KQL or Lucene to detect issues across indices. Selected

Machine Learning — Select ML job to detect anomalous activity. Select

Threshold — Aggregate query results to detect when number of matches exceeds threshold. Select

Event Correlation — Use Event Query Language (EQL) to match events, generate sequences, and stack data. Select

Indicator Match — Use indicators from intelligence sources to detect matching events and alerts. Select

New Terms — Find documents with values appearing for the first time. Select

ES|QL TECHNICAL PREVIEW — Use Elasticsearch Query Language (ES|QL) to find events and aggregate search results. Select

Source

Use Kibana Data Views or specify individual index patterns as your rule's data source to be searched.

Index Patterns Data View

apm-*-transaction* × auditbeat-* × endgame-* × filebeat-* × logs-* × packetbeat-* × traces-apm* × winlogbeat-* × -*elastic-cloud-logs-* ×

Enter the pattern of Elasticsearch indices where you would like this rule to run. By default, these will include index patterns defined in Security Solution advanced settings.

Custom query Import query from saved timeline

event.category : "malware"

Rule preview

Rule preview reflects the current configuration of your rule settings and exceptions, click refresh icon to see the updated preview.

Select a preview timeframe

Last 1 hour Refresh

Rule Preview

Count

file 0
intrusion_det... 0
malware 0

Note: Alerts with multiple event.category values will be counted more than once.

Columns Sort fields 4 alerts Fields

Acti...	@timestamp	Rule	Assignees
	2023-12-20T20:19:57....	Preview Rule	—
	2023-12-20T20:04:57....	Preview Rule	—
	2023-12-20T19:54:57....	Preview Rule	—
	2023-12-20T19:39:57....	Preview Rule	—

elastic

Search Elastic

Machine Learning > Anomaly Detection > Create job > Multi-metric

Create job: Multi-metric

Using index pattern gallery-*

1 Time range — 2 Pick fields — 3 Job detai

New job from index pattern gallery-*

Data split by method

TRACE

GET

High mean(bytes)

35000 30000 25000 20000 15000 10000 5000 0

09-01 00:00 10-01 00:00 11-01 00:00 12-01 00:00 01-01 00:00 02-01 00:00 03-01 00:0

Job ID
ml-gallery-01

Bucket span
15m

Job description
No description provided

Split field
method

Groups
No groups selected

Influencers
method, clientip

Start immediately

If unselected, job can be started later from the jobs list.

View results | Reset job | Start job running in real time | Create alert rule

Create rule

Name

Name is required.

Tags (optional)

Check every 1 minute

Notify Only on status change

Anomaly detection alert

Alert when anomaly detection jobs results match the condition. Documentation

BETA

Select job
ml-gallery-01

Result type

Bucket	Record	Influencer
How unusual was the job within the bucket of time?	What individual anomalies are present in a time range?	What are the most unusual entities in a time range?
Selected	Select	Select

Severity 75

0 25 50 75 100

Include interim results

Advanced settings

Check the rule condition with an interval

15d, 6m

Test

Actions

Select a connector type

Email | IBM Resilient | Index | Jira | Microsoft Teams

Cancel | Save

SIEM / Detections / Signals (SIEM Detections)

Search

+ Add filter

Signals (SIEM Detections) | External alerts

Signal count

Showing: 8 signals

8 7 6 5 4 3 2 1 0

02-19 11:35 02-19 11:40 02-19 11:45

Signals

View in timeline

Selected 0 signals | Close selected | Select all 8 signals

@timestamp	Rule	Version
Feb 19, 2020 @ 11:58:15.518	Telnet Port Activity with ti...	4
Feb 19, 2020 @ 11:58:15.517	Telnet Port Activity with ti...	4
Feb 19, 2020 @ 11:58:15.517	Telnet Port Activity with ti...	4

Telnet Port Attacks | Telnet Port hunter | Notes 0 | Feb 19, 2020 @ 11:5 → Fel

OR network.transport: "tcp" AND destination.port: "23"

Drop here to build an OR query

AND Filter | Search | KQL | All events

+ Add filter

All events | Raw events | Signal events

Columns | @timestamp | message | event.category

Feb 19, 2020 @ 11:58:15.518 — network_traffic

Feb 19, 2020 @ 11:58:15.517 — network_traffic

Feb 19, 2020 @ 11:58:15.517 — —

C4oC93AnvKHubXau6 | connection | {"msg":"Connection attempt rejected.","code":"REJ"} | Sr

467062ns

inbound | 80B | 2 pkts | tcp | 1:ogjHgVhE1+KJNtCSUlgqJ4KAbOk=

Source [redacted] : [redacted] — (50.00%) 40B 1 pkts → / ← (50.00%) 40B 1 pkts — Destination [redacted] : 23

Feb 19, 2020 @ 11:58:15.517 — —

CGBqNE4qdKn4kVs828 | connection | {"msg":"Connection attempt rejected.","code":"REJ"} | Sr

286102ns

inbound | 80B | 2 pkts | tcp | 1:Zq2X+mEhmnS33qukhi/BPdbgRYM=

Source [redacted] : 19153 — (50.00%) 40B 1 pkts → / ← (50.00%) 40B 1 pkts — Destination [redacted] : 23